To address this let us revisit our graphical illustration of market impact using the supply-demand curves. When a new buyer enters the market the demand curve shifts out while the supply curve shifts up to account for order information. A new clearing price P_2 emerges at the intersection of these new curves, and is determined from the slope of the supply and demand curves. This slope happens to be the price elasticity of demand and supply. In actuality, it is often difficult to ascertain the correct price elasticity for a physical good, and even more difficult to ascertain the correct price elasticity for a financial instrument. However, the volatility term serves as an effective proxy for the financial instruments price elasticity term. Notice that volatility is present in each of the variations of I-Star above. Volatility is used in the model to assist us uncover how market impact will differ across stocks. This is explained as follows:

The instantaneous impact equation for a particular stock k is:

$$I_k^* = a_1 \cdot \left(\frac{Q_k}{ADV_k}\right)^{a_2} \cdot \sigma_k^{a_3} \tag{4.17}$$

Rewrite this expression as follows:

$$I_k^* = \underbrace{\{a_1 \cdot \sigma_k^{a_3}\}}_{Sensitivity} \cdot \underbrace{\left(\frac{Q_k}{ADV_k}\right)^{a_2}}_{Shape} \tag{4.18}$$

We now have a sensitivity expression $a_1 \cdot \sigma_k^{a_3}$ which is stock specific and a shape expression $\left(\frac{Q}{ADV}\right)^{a_2}$ which is a universal shape relationship across all stocks. If we have parameter $a_2 = 1$ then we have a linear function where its slope is $a_1 \cdot \sigma_k^{a_3}$. This is identical to the supply-demand representation we showed above. In our formulation, we allow for non-linear supply and demand curves where each stock has its own sensitivity but the shape of the curve is the same across all instruments (which has been found to be a reasonable relationship).

A natural question is why do we not estimate these parameters at the stock level? The answer—we do not have sufficient data to estimate these parameters for all stocks. If we look at market impact for a single stock, change is often dominated by market movement and noise making it very difficult to determine robust and stable parameters at the stock level. In the next chapter, we show challenges behind fitting a stock level model.

Comparison of Approaches

How do the Almgren & Chriss and I-Star models compare? Readers might be interested to know that both of these models will converge to